

VIII. IMPLEMENTATION CONTRACTS AND EDGE SEMANTICS

Reproducibility requires more than a high-level algorithm name. Each public class validates two-dimensional finite input, stores the observed feature count, preserves original class labels, and raises an explicit pre-fit error. The tree exposes depth and leaf count as properties and importance as the method required by the brief; the forest exposes aggregate importance as a property. Predictions always return the original label dtype. Probabilities follow sorted global class order and sum to one, including when a bootstrap tree observes only a strict subset of classes.

A. Tree Search Invariants

At a node containing n_v rows and p_v candidate features, stable sorting costs $O(p_v n_v \log n_v)$ and a complete threshold scan costs $O(p_v n_v K)$ only for the vector impurity arithmetic. Empty-weight children are rejected even if they contain zero-weight rows. This distinction matters: sample count governs structural stopping, while total positive weight governs the split objective and leaf distribution. Multiplying every sample weight by a positive constant cannot change a split.

The test matrix checks one feature, all one class, noncontiguous and string labels, repeated values, constant features, deterministic equal-gain features, zero-weight observations, entropy, feature subsampling, and root depth zero. XOR is retained as a conceptual defense case. A greedy impurity tree correctly halts because neither root feature has positive gain; depth alone does not add look-ahead. The limitation helps explain why additive stumps can create a more expressive boundary over rounds even though one stump cannot solve the task.

B. Ensemble State and Numerical Safety

AdaBoost stores only accepted estimators. Its estimator error array contains the unclipped observed errors, while clipping is restricted to logarithms and weight updates. A first learner at or beyond the class-aware random threshold raises a clear error; a later rejected learner ends the sequence and leaves the previous valid ensemble usable. For SAMME.R, probability floor 10^{-12} prevents $\log 0$ and centered log-probabilities prevent a common offset from changing class preference. Stable softmax subtracts each row maximum.

Forest bootstrap and tree seeds come from independent children of one `SeedSequence`; workers never inherit mutable RNG state. The complete bootstrap array and its exact OOB complement are retained. A forest with OOB scoring disabled does not silently expose a meaningless score, and requesting OOB with `bootstrap=False` is invalid. Prediction ties resolve through global class order, matching the deterministic tree rule.

C. Clustering Details

K-Means final inertia is recomputed after the final centroid update, preventing a mismatch between stored labels, centroids, and objective. When duplicate points make a cluster

TABLE VI
BINDING INTERFACES AND PRINCIPAL FITTED EVIDENCE.

Class	Principal fitted state
DecisionTree	<code>classes_</code> , <code>root</code> , <code>depth</code> , <code>leaves</code> , <code>importances</code>
AdaBoost	<code>stumps</code> , <code>estimator weights/errors</code> , <code>stages</code>
Random Forest	<code>trees</code> , <code>bootstrap/OOB</code> <code>indices</code> , <code>coverage/score</code>
PCA	<code>training mean</code> , <code>ordered components</code> , <code>variance ratios</code>
K-Means	<code>centroids</code> , <code>labels</code> , <code>inertia</code> , <code>iteration count</code>
DBSCAN	<code>labels</code> , <code>core-sample indices</code>

empty, the row with largest current residual is selected deterministically and cannot be reused for two simultaneous empty clusters. DBSCAN precomputes inclusive squared neighborhoods. Expansion uses a queue and a membership flag so duplicate neighbor discoveries do not cause unbounded work. A row marked noise before another core point is expanded can be relabeled as border, matching reachability semantics.

IX. COMPLETE EXPERIMENT EVIDENCE CONTRACT

Each required experiment preserves raw units—stage, fold, replicate, depth, bootstrap model, or cluster configuration—before a plot or summary is made. This prevents a polished average from becoming the only surviving evidence.

A. Hypotheses Before Observing Results

The experiments were designed around falsifiable expectations rather than a post-hoc search for favorable plots. Increasing AdaBoost stages should first reduce training error monotonically or nearly so; test error may bottom out if later stumps focus on ambiguous samples. Increasing forest size should reduce the variability of test and OOB estimates, with diminishing returns once the correlated component in (8) dominates. Increasing depth should reduce bias but increase individual-tree variance; bootstrap averaging may tolerate deeper trees than a single-tree baseline.

Noise creates a mechanism-specific contrast. AdaBoost can amplify mislabeled rows through repeated error weighting. A forest never shares sample weights across trees and therefore distributes the damage over bootstrap samples, but deep trees can still memorize corrupt labels. Consequently, “forest is always more robust” is not treated as a theorem; replicate uncertainty and data geometry determine the observed conclusion.

Severe imbalance poses a different risk. Accuracy can remain high while a rare class is never predicted, so macro F1 and macro OVR AUC are mandatory. Training- fold oversampling improves the chance that rare classes influence a split, but can increase variance through repeated rows. Because the test distribution is untouched, the evaluation remains a test of population-like performance rather than an artificially balanced target.

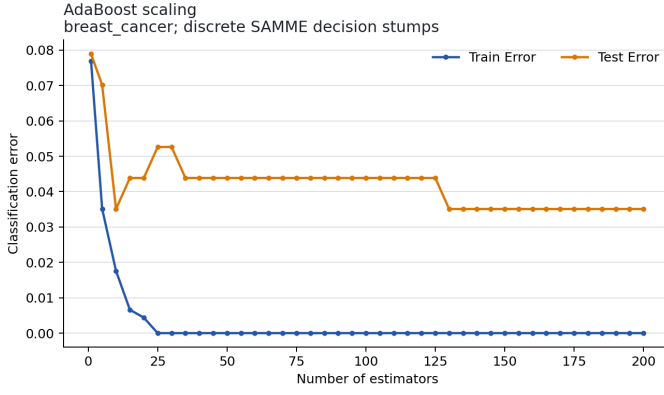


Fig. 5. Full staged AdaBoost curve through 200 accepted rounds unless early stopping occurs.

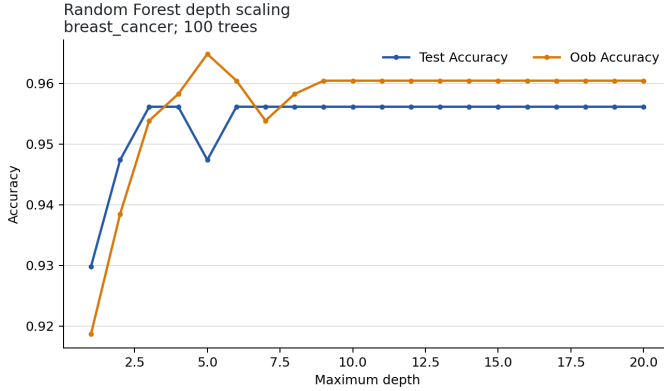


Fig. 6. Full depth sweep with separate OOB and clean-test accuracy.

B. Complexity and Resource Strategy

The implementation intentionally prioritizes clarity, but avoids the most expensive naive operations. Tree class counts are incremental; staged AdaBoost predictions reuse one maximum fit; Random Forest size curves reuse prefixes of one 200-tree forest; distances used by DBSCAN and the k -distance plot are computed only on a deterministic visualization sample. Bias-variance probabilities are compressed to NPZ rather than serialized model by model. The completed full run required 6.88 hours. Its manifest, generated artifacts, and validation trail retain the exact configuration and numerical evidence needed for audit without changing model seeds or metrics.

X. BONUSES, THREATS TO VALIDITY, AND DEFENSE READINESS

A. Gradient Boosting Extension

The bonus classifier uses binary labels $y_i \in \{0, 1\}$ and an initial raw score $F_0 = \log[\bar{y}/(1-\bar{y})]$. At stage m , probabilities are $p_i = \sigma(F_{m-1}(x_i))$ and the negative log-loss gradients are

$$r_{im} = y_i - p_i. \quad (11)$$

A shallow from-scratch regression tree fits r_{im} by midpoint splits that minimize child squared error. The update $F_m(x) = F_{m-1}(x) + \eta h_m(x)$ uses a stable sigmoid for both signs of F .

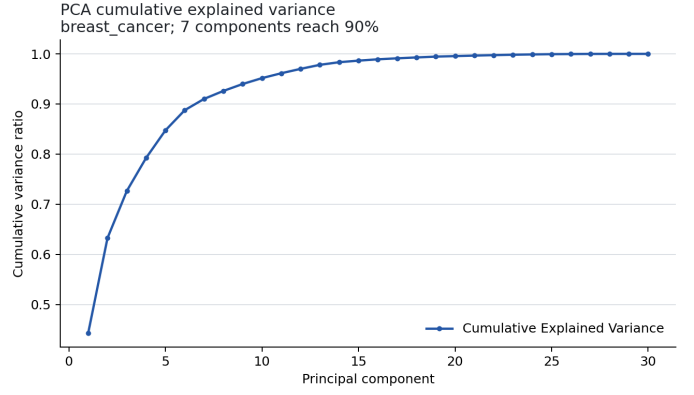


Fig. 7. Cumulative variance provides a reproducible dimensionality rule.

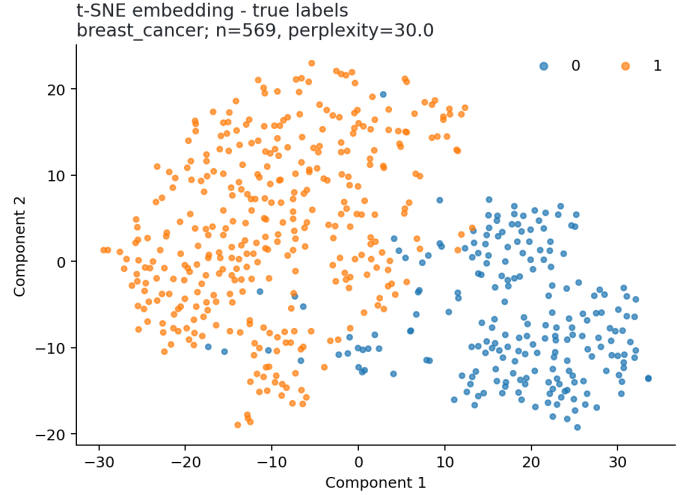


Fig. 8. t-SNE is a local visualization bonus and is not used to assert global metric cluster structure.

Training log loss is stored after every stage, and tests require a decreasing trend, finite normalized probabilities, and equality between the last staged prediction and the final prediction. The method is intentionally binary; a multiclass extension is outside the claimed bonus scope.

B. PCA Versus t-SNE

PCA is deterministic, linear, and provides explained-variance accounting. t-SNE uses a fixed stratified sample, PCA initialization, automatic learning rate policy, recorded perplexity, and seed 42. It supplies a complementary local-neighborhood view, not a clustering score. Apparent separation can vary with perplexity and random optimization; distant point-to-point distances, cluster area, and relative inter-group spacing are therefore excluded from the argument. K-Means and DBSCAN remain evaluated in the retained PCA space.

C. Threats to Internal and External Validity

Reference agreement establishes a useful implementation check but does not prove identical behavior: tie rules, stopping tolerances, category treatment, and probability conventions

can differ. To reduce this threat, correctness is also tested with hand-computed cases and invariants. Fixed seeds improve audit repeatability but do not represent uncertainty; cross-validation, five noise replicates, and 100 bootstrap replicates supply the required variation.

Dataset conclusions are not universal. Breast Cancer is small and mostly continuous; Adult has mixed socioeconomic features and missing categories; Covertypes is larger, multiclass, high-dimensional, and severely imbalanced. The three together vary important factors, but they do not span concept drift, adversarial noise, image structure, or calibrated-risk applications. The project therefore reports conditional findings tied to source versions and realized samples.

Bias–variance estimates use a finite fixed test set. They answer how fitted probabilities vary across bootstrap training samples at those points, not the entire population expectation. The reported decomposition residual checks algebra, while it cannot remove sampling uncertainty. Similarly, OOB accuracy is based only on covered rows and is not identical to five-fold CV.

D. Defense and Release Gates

Each primary module has a different reviewer and a backup owner. Defense drills cover the derivation and code location of weighted impurity, SAMME’s $\log(K - 1)$ term and rejection threshold, the correlation term in forest variance, OOB coverage, PCA centering, K-Means empty clusters, and DBSCAN’s noise-to-border transition. A member claiming work must substantiate it with a real commit, review, experiment run, report edit, or slide contribution.

The release gate requires all tests, at least 85% measured coverage, clean Ruff and mypy output, a successful no-argument full experiment run, dataset constraint assertions, a valid executed notebook, six content pages plus references, a 10–12 slide rendered deck, an all-member-approved contribution report, and an auditable Git history. Only then is `v1.0-final` created and the exact tagged artifacts uploaded to Moodle.